🩺 CardioCare AI Portal: End-to-End System Architecture

Autonomous Rural Telemedicine & Cardiac Triage Kiosk | KIIT Deemed to be University (School of Computer Engineering)

🖥️ Kiosk Edge Client

Next.js 16 + React 19

- **Wake Word Voice AI:** Continuous listening via Web Speech API ("Hello Doctor").
- **Live Telemetry:** Web Bluetooth API connects to consumer smartwatches (GATT 0x180D).
- **Real-time EKG:** HTML5 Canvas 60fps dynamic cardiac waveform renderer.
- **Prescription OCR:** Client-side Tesseract.js WebAssembly text extraction.

⚙️ Gateway Backend

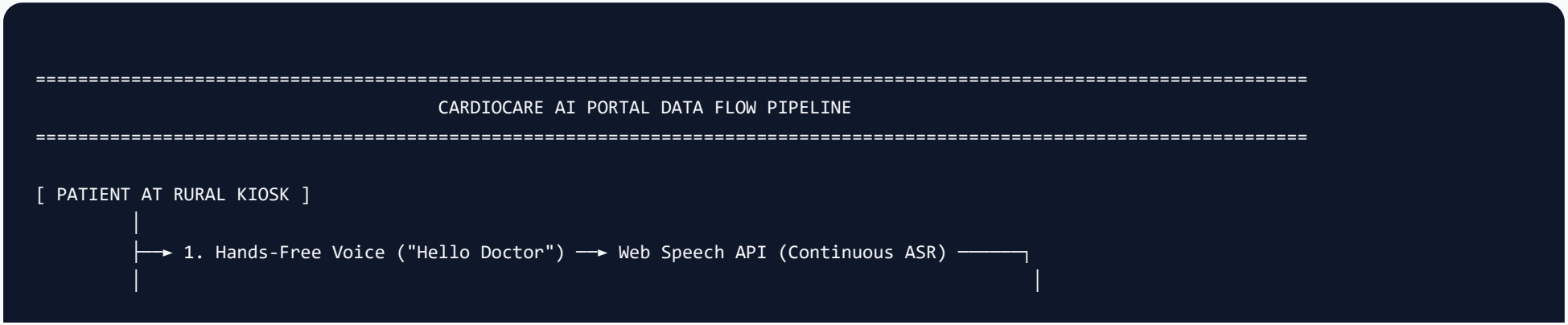
FastAPI + Python 3.10

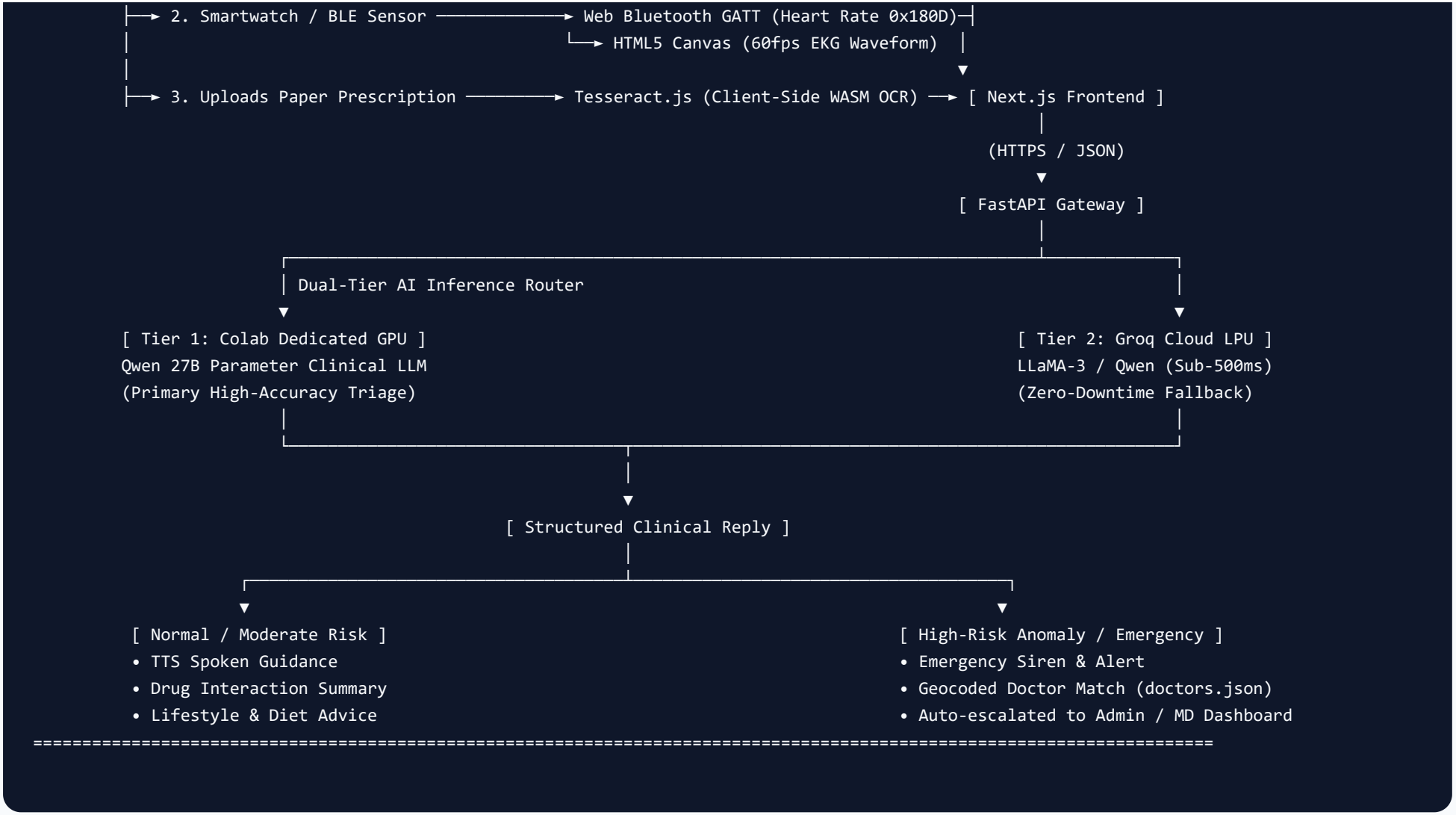
- **Async REST Router:** High-throughput ASGI server via Uvicorn.
- **Intelligent Prompt Engine:** Clinical cardiology context injection and markdown hygiene.
- **Resilient Dual-Inference:** Primary Colab GPU with automated Groq Cloud LPU fallback.
- **Edge Persistence:** Flat-file JSON database (`users.json`, `doctors.json`) for rural kiosks.

🧠 AI & Model Core

Qwen 27B + Groq LPU

- **Primary Model:** 27B-parameter Qwen clinical LLM hosted on remote Colab GPU via ngrok.
- **High-Speed Fallback:** Groq Cloud LPU running LLaMA-3 / Qwen (sub-500ms response).
- **Vision/OCR:** Tesseract.js client OCR with rule-based medication artifact cleanser.
- **Emergency Escalation:** Deterministic risk classifier routing severe vitals to human cardiologists.





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